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# SpikeProphecy: A Large-Scale Benchmark for Autoregressive Neural Population Forecasting

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## Abstract

Neural population models, which predict the joint firing of many simultaneously recorded neurons forward in time, are typically evaluated by a single aggregate Pearson correlation  $r$  between predicted and actual spike counts, a number that masks critical structure. We argue that how we evaluate spike forecasting matters as much as what we build, and introduce SpikeProphecy, the first large-scale benchmark for causal, autoregressive spike-count forecasting on real electrophysiology recordings. Our core contribution is a population metric decomposition that separates aggregate performance into temporal fidelity, spatial pattern accuracy, and magnitude-invariant alignment. The decomposition surfaces aspects of the underlying data that an aggregate scalar collapses together. We apply the protocol to 105 Neuropixels sessions (Steinmetz 2019 + IBL Repeated Site;  $\sim 89,800$  neurons) with seven architecture baselines spanning four structural families: four SSMS (three diagonal and one non-diagonal), a Transformer, an LSTM, and a spiking network. The decomposition surfaces a brain-region predictability ranking that reproduces across all seven baselines and survives ANCOVA correction for firing-statistics constraints (region  $\Delta R^2=0.018$  above the firing-statistics covariates). It also exposes a sub-Poisson evaluation floor where rigorous metrics combine with genuine biophysical constraints on regular spike trains, and yields a negative result on KL-on-output-rates distillation for ANN $\rightarrow$ SNN transfer in this Poisson count domain.

# 1 Introduction

High-density electrode probes such as Neuropixels record the simultaneous spiking activity of hundreds to thousands of neurons in behaving animals. A growing class of sequence models, which we collectively call *neural population models*, has been trained on these recordings using modern architectures borrowed from language and time-series work [Pandarinath et al., 2018, Ye and Pandarinath, 2021, Azabou et al., 2023]. These models are overwhelmingly evaluated on a single downstream task, *behavioral decoding*: mapping spikes to cursor velocity, stimulus identity, or movement direction. The more fundamental computational challenge is *spike forecasting*: predicting the future firing of thousands of neurons from their own recent history. This task is critical for closed-loop brain-computer interfaces (BCIs), where 50–100 ms look-ahead predictions compensate for sensing and processing delays, and for *in silico* neural population simulators (“digital twins”) that could accelerate BCI algorithm development without animal experimentation. Despite this importance, **no established benchmark exists** for spike-count forecasting at scale on real electrophysiology data.

Standard evaluation practices compound this gap. The community relies on aggregate per-neuron Pearson  $r$  as the primary metric, but a single number masks critical structure: brain region differences in predictability, neuron subpopulation failures, and the distinction between temporal dynamics capture and spatial pattern fidelity. An aggregate  $r = 0.50$  sounds reasonable, but decomposing it reveals that temporal population dynamics are well-captured ( $r_{\text{pop}} = 0.76$ ) while individual neuron spatial identity is only moderately captured ( $r_{\text{spatial}} = 0.55$ ). We need *population-level metrics* that measure what actually matters for downstream applications.

We introduce the **SpikeProphecy benchmark**, designed to address both gaps simultaneously. Our contributions are:

1. **Evaluation protocol (primary contribution).** A population-metric decomposition (`pop_rate_r` for temporal fidelity, `spatial_r` for spatial pattern, `cosine_sim` for magnitude-invariant alignment) that exposes structure invisible to aggregate Pearson  $r$ . The decomposition re-orders the brain-region predictability hierarchy (§4.1), exposes a sub-Poisson evaluation floor (§4.1), and separates the failure modes of the linear-vs.-deep modeling hierarchy (§4.2) that are all missed by single- scalar reporting. We argue the decomposition should become standard for neural population forecasting (§3.3).
2. **Task & data.** The first large-scale autoregressive spike-count forecasting benchmark: 105 sessions from two public Neuropixels datasets (~89,800 neurons), with standardized preprocessing, temporal splits, and a 14-test data-integrity audit suite covering 5 leakage vectors.
3. **Seven evaluated architectures.** Four SSMs spanning the diagonal/non-diagonal axis (Mamba, HGRN2, LRU: diagonal; GatedDeltaNet: non-diagonal), a Transformer, an LSTM, and a recurrent spiking network (RSynaptic SNN), all trained under identical optimizer, schedule, loss, and data so that decomposition findings can be attributed to the task and data rather than to any single architecture choice. Per-architecture results are reported in §4 and Table 4.
4. **Findings enabled by the benchmark.** The protocol’s stratified reporting surfaces structure that standard aggregate metrics smooth over: a brain-region predictability ranking that survives ANCOVA correction (region  $\Delta R^2=0.018$  above firing-statistics covariates), a sub-Poisson evaluation floor that combines biophysical hardness with metric harshness, and a controlled negative result for KL-on-output-rates distillation in this Poisson count setting (we do not claim the result generalizes to feature-level or attention-transfer distillation).
5. **Public ecosystem.** Processed tensors and preprocessing code (the source recordings remain at their public Figshare/IBL repositories), `pip`-installable evaluation toolkit, trained checkpoints, and YAML reproduction configs.

## 2 Related work

**Neural population models.** LFADS [Pandarinath et al., 2018] uses a sequential VAE to estimate firing rates but targets latent dynamics inference, not forecasting. NDT/NDT2 [Ye

and Pandarinath, 2021] applies masked attention to neuronal spiking for behavioral decoding. BRAID [Vahidi et al., 2025] is closest to our work (input-driven RNNs with a forecasting objective on Steinmetz recordings) but provides no multi-architecture comparison, no cross-dataset evaluation, and no population-level metrics. POCO [Duan et al., 2025] addresses neural forecasting but on calcium imaging (continuous fluorescence), not discrete spike counts, making it a fundamentally different data modality.

**Why not foundation models as baselines?** Recent neural foundation models such as POYO/POYO-2, NDT-2, and brain2vec are not direct comparisons to SpikeProphecy, and fair inclusion would require substantial adaptation rather than a drop-in evaluation. POYO/POYO-2 [Azabou et al., 2023] tokenize individual spike events and pretrain for *masked-token prediction* across heterogeneous sessions; adapting POYO to produce a causal, 50 ms-bin rate forecast requires swapping its output head and reintroducing a temporal causality constraint that its bidirectional attention violates. NDT-2 [Ye and Pandarinath, 2021] targets *latent inference* (unsupervised smoothing of current-bin rates under co-smoothing) and is evaluated primarily on behavioral decoding; forecasting the next bin from history alone is not a supported task in the public release. brain2vec and similar representation-learning approaches produce session-level embeddings rather than per-bin rate predictions. A comparison that required retraining each of these under our causal forecasting task with matched compute and data would itself be a paper, and we acknowledge that this absence is a real gap rather than a methodological choice: without the current generation of neural foundation models, our architecture-level findings (§4.2) cannot claim to reflect the state of the art. We release our exact causal dataloaders and temporal splits so the community can adapt and benchmark foundation models on strict causal forecasting using the same substrate.

**Neural data benchmarks.** The Neural Latents Benchmark (NLB; Pei et al. 2021) standardized evaluation of latent variable models using co-smoothing bits per spike (co-BPS) on 4 motor/cognitive datasets. NLB targets *latent inference* (unsupervised rate smoothing), whereas SpikeProphecy targets *causal forecasting* of future spike counts from history alone. NeuroBench [Yik et al., 2025] benchmarks neuromorphic algorithms but on classification tasks (keyword spotting, event vision), not neural time-series. SpikeProphecy bridges this gap: the first benchmark for autoregressive spike prediction at scale, with evaluation metrics that decompose performance beyond scalar summaries. Table 1 contrasts our evaluation protocol with existing practice.

**Sequence architectures for neural data.** Structured state-space models (S4, S5) and their successors (Mamba [Gu and Dao, 2023], LRU [Orvieto et al., 2023]) have shown strong performance on long-range sequence tasks. We provide the first systematic comparison of these architectures on neural spike data, including evaluation beyond Pearson  $r$ .

**Evaluation methodology.** Standard neural data metrics (per-neuron  $r$ , co-BPS,  $R^2$ ) are all aggregate scalar summaries that collapse temporal and spatial fidelity into a single number. HELM [Liang et al., 2023] demonstrated in the LLM domain that multi-axis evaluation (accuracy  $\times$  fairness  $\times$  calibration) reveals trade-offs invisible to single-metric leaderboards. Our population-level metrics bring this philosophy to neuroscience, inspired by population vector analysis [Georgopoulos et al., 1986] but applied to forecast evaluation.

**ANN-to-SNN distillation.** Knowledge distillation for spiking networks [Hong et al., 2023] has been studied exclusively on image classification (CIFAR, ImageNet). No prior work evaluates ANN-to-SNN distillation on spike forecasting; our negative result is the first empirical demonstration, with a hypothesized mechanism (the redundancy of soft labels when the target is already real-valued), for why standard KL-weighted distillation fails in Poisson regression domains.

### 3 Benchmark design

#### 3.1 Task formulation

Given a history window of  $T$  spike-count vectors, predict the next time bin:

$$X_t = \{\mathbf{x}(t-T+1), \dots, \mathbf{x}(t)\} \longrightarrow \hat{\mathbf{y}}(t+1) \approx \mathbf{x}(t+1), \quad \mathbf{x}(t) \in \mathbb{Z}_{\geq 0}^M \quad (1)$$

where  $M$  is the number of simultaneously recorded neurons in a session (varies between sessions, up to  $M_{\max}=1,998$  in the largest IBL session),  $\Delta t = 50$  ms, and  $T = 10$  bins (500 ms history).  $M_{\max}$  is a property of the data (we pad shorter sessions to it and mask the unused channels; see below) rather than a model hyperparameter we tune; the architectures we evaluate also support per-session input/output projections that side-step the shared- $M_{\max}$  design, but the results in this paper use the shared projection for parameter parity. The model outputs predicted rates  $\hat{\lambda}_i > 0$  via softplus, trained with Poisson NLL. Two constraints define the task: it is *strictly autoregressive* (only intrinsic covariates, i.e. past spikes, are used; no stimulus or behavioral features) and *causal* (no future context at any layer, including attention). Autoregressive implies causal, but causal does not imply autoregressive (a model can be causal with external inputs); we require both. This rules out stimulus information that would not be available at BCI inference time and rules out bidirectional attention used in masked-token foundation models such as POYO/NDT-2.

For multi-session training, spike counts are zero-padded to  $M_{\max}$  with a per-sample binary channel mask; the loss operates only on real (unmasked) channels.

#### 3.2 Datasets

**Steinmetz 2019 (39 sessions).** Ten mice, 39 recording sessions using Neuropixels probes spanning visual cortex, motor cortex, hippocampus, thalamus, and midbrain [Steinmetz et al., 2019]. Up to 1,240 simultaneously recorded neurons per session. Visual discrimination task ( $\sim 2$ -hour recordings). Single lab, single rig, controlled conditions. Source data are publicly available on Figshare under CC-BY-4.0 (<https://doi.org/10.6084/m9.figshare.9598406.v2>); we redistribute only our processed binned and split tensors and the code that derives them.

**IBL Repeated Site (66 sessions).** Multi-lab international consortium, standardized probe trajectory across 9 labs [IBL et al., 2024]. Up to 1,998 simultaneously recorded neurons. Same task paradigm across different labs, mice, and rigs; tests cross-lab generalization. Source data are publicly available via the IBL Open Neurophysiology Environment ONE API (<https://www.internationalbrainlab.com/data>); as above, we redistribute only the processed tensors and the preprocessing pipeline.

**Processing and splits.** Raw spike times are binned at 50 ms into integer count vectors. Each session is split temporally into three contiguous blocks (70/15/15 train/val/test, ordered first/middle/last in time) to prevent information leakage. Interleaved or random splits would introduce trivial leakage on autoregressive forecasting because adjacent 50 ms bins are temporally correlated.

**Auditable leakage suite.** We ship a 14-test automated audit suite that runs on every commit and verifies five concrete leakage vectors: train/test bin overlap, sliding-window boundary crossing, cross-session spillover, normalization using future statistics, and history-feature leakage. The population GLM result reported in Table 4 ( $r=1.000$  on train,  $r=-0.015$  on val) is the canonical example of what these tests are designed to catch when a benchmark grants linear models excessive feature access on a temporally-split task. Temporal-split leakage tests are not novel in themselves; we contribute the packaged suite plus the canonical Population-GLM catch as a reusable artifact that runs on every commit. At this scale, leakage failures are statistically indistinguishable from genuine signal at  $r \sim 0.1$ , and we expect the suite to be more useful than the specific tests it currently contains.

### 3.3 Evaluation protocol: population metric decomposition

*This subsection describes our primary contribution.* Each of the three component metrics below is individually well-precedented in computational neuroscience (population vector analysis [Georgopoulos et al., 1986], per-bin cross-neuron correlation, magnitude-invariant cosine alignment). Our contribution is to standardize this triplet as the reporting protocol for spike-count forecasting, paired with stratified reporting by brain region and Fano factor, and to demonstrate empirically (§4.1) that the protocol surfaces structure that aggregate  $r$  hides.

$r_{\text{pop}}$ ,  $r_{\text{spatial}}$ , and  $\text{cos\_sim}$  are three independent summary statistics that describe distinct axes of fit quality (one marginalizes neurons before correlation, one marginalizes time, one removes magnitude); they are not orthogonal components of an algebraic identity that sums to  $\text{Wt-}r$ . We define each metric formally, state what neural phenomenon it targets, and compare the resulting protocol to existing practice (Table 1).

**Standard metrics (baseline protocol).** Existing neural population benchmarks rely on: (1) Aggregate Pearson  $r$  across neurons  $\times$  time bins (*weighted Pearson  $r$* , abbreviated  $\text{Wt-}r$ ); (2) per-neuron median  $r$ ; (3) co-smoothing bits per spike (co-BPS); (4) Poisson NLL; (5) mean absolute error (MAE). All five are *scalar summaries* that conflate temporal and spatial fidelity into a single number. We define  $\text{Wt-}r$  explicitly as the activity-weighted mean of per-neuron Pearson correlations:

$$\text{Wt-}r = \frac{\sum_{i=1}^M w_i \cdot \text{Pearson}(y_i, \hat{y}_i)}{\sum_{i=1}^M w_i}, \quad w_i = \text{Var}(y_i), \quad (2)$$

where the weighting downweights near-silent neurons whose correlation is dominated by zeros. This is the headline metric reported in Table 4; it is reported here for backwards comparability, not as the contribution.

**Population metric decomposition (our contribution).** Let  $\mathbf{y}(t), \hat{\mathbf{y}}(t) \in \mathbb{R}^M$  denote the ground-truth and predicted activity vectors at time bin  $t$ , across  $M$  neurons and  $T_{\text{eval}}$  evaluation bins.

**(i) Population Rate  $r$  (pop\_rate\_r).** Temporal fidelity: *does the model capture when the population is active?*

$$r_{\text{pop}} = \text{Pearson}\left(\left[\sum_{i=1}^M y_i(t)\right]_{t=1}^T, \left[\sum_{i=1}^M \hat{y}_i(t)\right]_{t=1}^T\right) \quad (3)$$

This marginalizes over neuron identity to measure whether the model tracks the ensemble firing rate envelope.

**(ii) Spatial Pattern  $r$  (spatial\_r).** Spatial fidelity: *does the model capture which neurons fire?*

$$r_{\text{spatial}} = \frac{1}{T_{\text{eval}}} \sum_{t=1}^{T_{\text{eval}}} \text{Pearson}(\mathbf{y}(t), \hat{\mathbf{y}}(t)) \quad (4)$$

This measures per-timebin cross-neuron correlation, capturing whether the model identifies the active subset of neurons at each moment.

**(iii) Population Cosine Similarity (cosine\_sim).** Magnitude-invariant pattern alignment: *does the model capture relative activation patterns regardless of overall rate?*

$$\text{cos\_sim} = \frac{1}{T_{\text{eval}}} \sum_{t=1}^{T_{\text{eval}}} \frac{\mathbf{y}(t) \cdot \hat{\mathbf{y}}(t)}{\|\mathbf{y}(t)\| \|\hat{\mathbf{y}}(t)\|} \quad (5)$$

By normalizing magnitudes, this metric isolates pattern fidelity from rate calibration; a model may correctly predict which neurons are active (high cosine) while over/under-predicting rates (low MAE). Because spike counts and softplus rate predictions are strictly non-negative, cosine similarity here is bounded inside the positive orthant and has a non-zero floor. This is not a flaw: a parameter-free “predict the train-set mean” baseline reaches  $\text{cos}=0.31$  (the

empirical positive-orthant floor on this task), the autoregressive GLM reaches  $\text{cos}=0.24$ , the modern-recurrence cluster reaches  $\text{cos}=0.61\text{--}0.63$ , and the trailing baselines (LSTM, SNN) reach  $\text{cos}\approx 0.58$  (Table 4). The full dynamic range (0.31–0.63) is comparable to that of pop-rate  $r$  (0–0.76) and spatial  $r$  (0–0.55), so the metric remains discriminative on the relevant scale.

**Decomposition in practice.** An aggregate weighted  $r = 0.50$  decomposes as:  $r_{\text{pop}} = 0.76$  (temporal dynamics well-captured),  $r_{\text{spatial}} = 0.55$  (spatial patterns moderately captured), per-neuron median  $r = 0.12$  (individual neurons noise-dominated). These three axes have distinct operational implications: a model with high  $r_{\text{pop}}$  but low  $r_{\text{spatial}}$  is useful for population rate tracking (e.g., seizure detection) but not for single-neuron stimulation or neuron-specific BCI decoding.

Table 1: Comparison of evaluation protocols for neural population models. Existing benchmarks use scalar summaries; SpikeProphecy decomposes performance into interpretable axes.

| Metric         | Temporal? | Spatial? | Used by     |
|----------------|-----------|----------|-------------|
| Aggregate $r$  | ✓         | ✓        | Most work   |
| Per-neuron $r$ | ✓         | –        | Some work   |
| co-BPS         | ✓         | –        | NLB         |
| $R^2$          | ✓         | ✓        | Some work   |
| pop_rate_r     | ✓         | –        | <b>Ours</b> |
| spatial_r      | –         | ✓        | <b>Ours</b> |
| cosine_sim     | –         | ✓        | <b>Ours</b> |

**Stratified evaluation.** Beyond the three metrics, we advocate reporting results *stratified* by brain region (§4.1) and Fano factor class (§4.1). These stratifications expose structure that aggregate metrics conceal and that is stable across the seven baselines we tested under our shared training recipe. We propose this practice as a reporting standard.

### 3.4 Architecture baselines

All architectures share the same input pipeline, loss function, optimizer, and training schedule (50 epochs, AdamW, cosine LR with warmup, seed=42). Table 2 summarizes them.

Table 2: Architecture baselines in the SpikeProphecy benchmark. All models use shared output heads and identical training hyperparameters.

| Architecture   | Type                         | Params      | Key property                    |
|----------------|------------------------------|-------------|---------------------------------|
| Mamba          | Diagonal selective SSM       | 1.95M       | Input-dependent gating, $O(T)$  |
| HGRN2          | Diagonal gated linear RNN    | 1.82M       | State expansion, $O(T)$         |
| LRU            | Diagonal linear recurrence   | 1.23M       | Ring eigenvalue init            |
| GatedDeltaNet  | Non-diagonal delta-rule SSM  | 1.43M       | Matrix state per head, $O(T)$   |
| Transformer    | Causal attention             | 2.22M       | Global context, $O(T^2)$        |
| LSTM           | Gated recurrence             | 2.22M       | Classical baseline              |
| SNN            | Spiking (RSynaptic)          | 965K        | Event-driven, neuromorphic (3L) |
| Autoreg GLM    | Poisson, own $T$ -step hist. | $\sim 10/N$ | No cross-neuron info            |
| Population GLM | Poisson, full $(T, M)$ hist. | $\sim 7K/N$ | Linear pop baseline             |

**Mamba** [Gu and Dao, 2023] extends the linear state-space model with selective, input-dependent parametrization ( $d_{\text{model}}=256$ ,  $n_{\text{layers}}=3$ ,  $d_{\text{state}}=16$ ).

**HGRN2** [Qin et al., 2024]: Hierarchically Gated Recurrent Network with state expansion. A diagonal gated linear RNN with data-dependent forget gates ( $d_{\text{model}}=256$ ,  $n_{\text{layers}}=3$ ,  $\text{expand\_ratio}=2$  so  $d_{\text{state}}=512$ ). Included specifically as a second diagonal SSM for comparison against Mamba.

**GatedDeltaNet** [Yang et al., 2024]: a non-diagonal modern SSM using the gated delta rule, with a matrix state per head that updates by an outer-product correction ( $d_{\text{model}}=256$ ,  $n_{\text{layers}}=3$ ,  $n_{\text{heads}}=4$ ,  $d_{\text{head}}=64$ ). Included as the non-diagonal control to test whether diagonal structure is load-bearing for neural-data forecasting.

**Transformer:** Causal encoder-only with strict autoregressive masking ( $d_{\text{model}}=256$ ,  $n_{\text{heads}}=8$ ,  $n_{\text{layers}}=3$ , pre-norm).

**LRU** [Orvieto et al., 2023]: 3-layer linear recurrent unit with ring eigenvalue initialization ( $|\lambda| \in [0.8, 0.99]$ ).

**LSTM:** Standard 3-layer LSTM ( $H=256$ , dropout 0.2).

**SNN:** Recurrent spiking neural network using `snnTorch` RSynaptic neurons (two-compartment with learnable decay  $\beta$ ). Trained standalone with Poisson NLL (no distillation).

**Autoregressive GLM:** L2-regularized Poisson regression where each neuron’s GLM uses only its own  $T$ -step spike history as features ( $\alpha=10^{-4}$ , one model per neuron per session). A controlled no-cross-neuron-information baseline.

**Population GLM:** L2-regularized Poisson regression where each neuron’s GLM uses the full flattened  $(T, M)$  population history ( $\sim 7K$  features per model,  $\alpha=10^{-4}$ ). Same independent-per-neuron fit procedure; demonstrates that naive linear access to population context is not sufficient when the number of features is much larger than the number of samples.

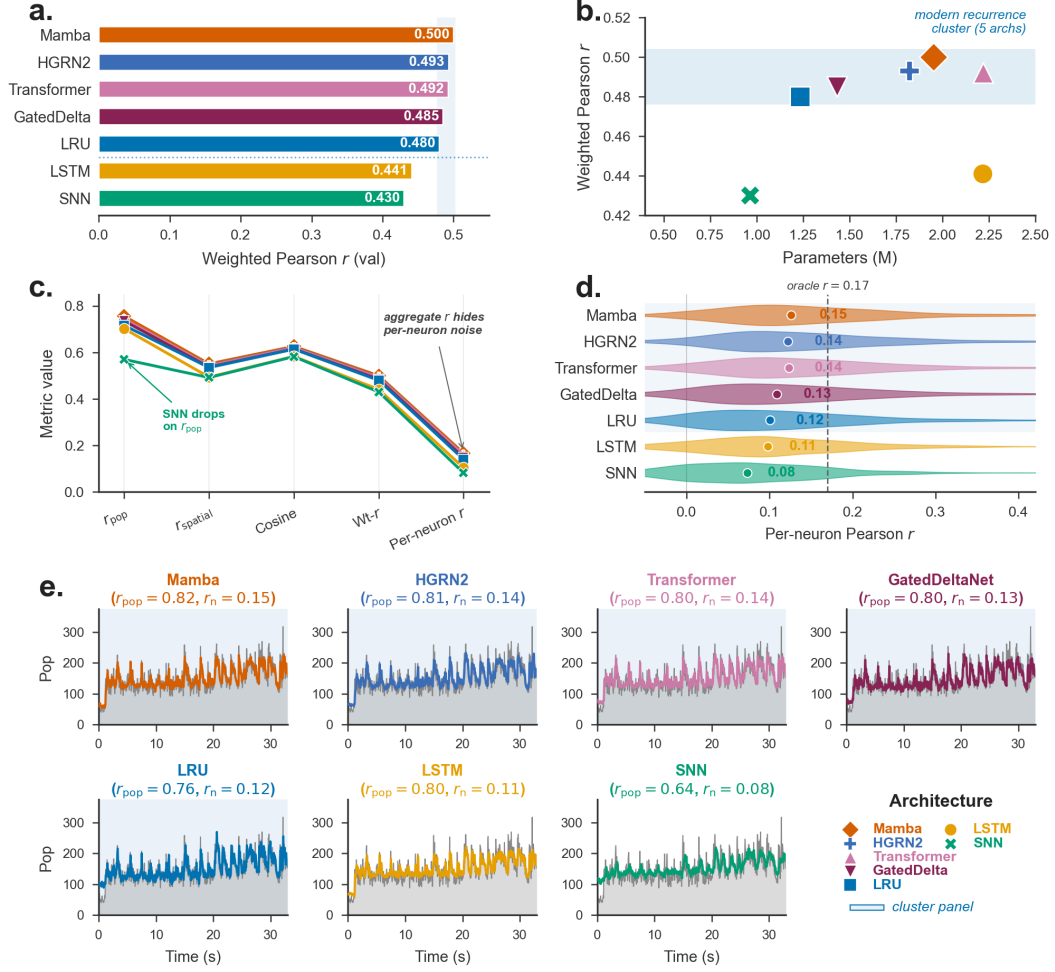
### 3.5 Benchmark access and usage

SpikeProphecy is designed as a **living benchmark ecosystem**, not a static dataset release. Usage follows three steps:

1. **Load data:** `ds = load_dataset("spikeprophecy/steinmetz")`
2. **Train and predict:** Use any architecture with the provided `DataLoader` and Poisson NLL loss.
3. **Evaluate:** `metrics = spikeprophecy.evaluate(pred, gt)`  $\rightarrow$  returns `pop_rate_r`, `spatial_r`, `cosine_sim`, stratified by region and Fano factor.

The evaluation toolkit outputs a standardized JSON report and optional radar plot for multi-axis comparison (Figure 1). Pre-computed baseline predictions for all seven architectures are included to enable comparison without retraining.

**Documentation methodology.** Beyond the evaluation protocol itself, we treat *how evaluative claims are documented* as a benchmark contribution. The release bundles two artifacts that make claims about this benchmark auditable: (i) a datasheet (Appendix A.9) covering motivation, composition, collection, preprocessing, intended use, and ethical considerations; and (ii) the leakage audit suite (§3.2) whose pass state on every commit is part of the released code. Together these let a downstream user verify, before running a single experiment, what the benchmark does and does not measure.



**Figure 1: SpikeProphecy benchmark overview.** Each architecture has its own color and marker, used consistently across panels (a)–(e). Panels (a)–(d) are the analytical summary across all 39 Steinmetz sessions and 27,212 neurons; panel (e) illustrates the predictions on a single near-median session. **(a)** Absolute weighted Pearson  $r$  per architecture (val split, sorted descending): the five cluster baselines fall in  $r=0.480-0.500$  (blue band), LSTM at 0.441, SNN at 0.430; the dotted divider tracks the Wilcoxon-significant gap between cluster and LSTM/SNN ( $p<10^{-7}$ , Appendix A.3). **(b)** Pareto: weighted Pearson  $r$  vs. parameter count; the five cluster baselines occupy a  $\sim 2$ -pp band ( $r=0.480-0.500$ , blue shaded region), LSTM and SNN visibly below. **(c)** Decomposition trace (parallel-coordinates) across five normalized metrics: cluster archs overlap on every axis, while SNN drops on  $r_{\text{pop}}$  and per-neuron  $r$  collapses an order of magnitude below the aggregate metrics for every architecture, the headline “aggregate  $r$  hides per-neuron noise” reveal. **(d)** Per-neuron  $r$  distribution per architecture (horizontal violins,  $\sim 700$  neurons of session 4 each). The empirical oracle ceiling ( $r=0.17$ , dashed) anchors the scale; cluster violins overlap, LSTM and SNN are visibly left-shifted. **(e)** Population-rate small multiples on a near-median Steinmetz session (session 4,  $N=703$ ,  $\sim 33$  s of 50 ms bins; per-arch  $r$  values within  $|\Delta| \leq 0.0075$  of the 39-session per-arch median). Cluster panels (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU) carry a faint blue background. Each title reports session-4  $r_{\text{pop}}$  (population rate) and  $r_n$  (per-neuron Pearson averaged over  $\sim 700$  neurons): pop-rate is similar across all seven baselines on this session, while per-neuron  $r$  falls almost  $2\times$  from cluster ( $\sim 0.13-0.15$ ) to SNN ( $\sim 0.08$ ).



## 4 Results

### 4.1 Decomposition reveals structure invisible to aggregate $r$

The population metric decomposition and stratified evaluation reveal two findings that are **invisible to standard aggregate metrics**. Each demonstrates structure that would be missed by reporting a single Pearson  $r$ .

**Finding 1: A functional brain-region predictability hierarchy.** Parsing Allen CCF brain region labels for 27,212 neurons across 39 sessions reveals a predictability hierarchy that remains significant after ANCOVA correction for log firing rate and Fano factor (Figure 2a). The full ANCOVA model ( $\text{model\_r} \sim \text{log\_rate} + \text{fano} + \text{region}$ ) reaches  $R^2=0.275$  at the 8-system grouping; the covariates alone reach  $R^2=0.257$ , so the *region-incremental* contribution beyond firing-statistics is  $\Delta R^2=0.018$  at this granularity. The increment is modest but reproducible: it survives ANCOVA correction (partial  $F>10^2$ ,  $p < 10^{-77}$ ) and is robust to grouping choice. At a coarser 4-class grouping (Cortex/Subcortex/Hippocampal/Other) the increment drops to  $\Delta R^2=0.011$ ; at the fine 54-region Allen acronym level it rises to  $\Delta R^2=0.053$  (Appendix A.5). As Figure 2a shows, the ranking itself is reproducible across all seven baselines: every architecture traces the same monotonic hierarchy across the eight functional regions despite their absolute-level differences, which is the more substantive finding than the absolute  $R^2$ . The uncorrected Kruskal–Wallis statistic ( $H=1,056$ ,  $p < 10^{-200}$ ) is inflated by the large neuron count and should not be interpreted as an effect size.

Table 3: **Finding 1: Brain-region predictability hierarchy** (Mamba per-neuron Pearson  $r$ , val split; 21,689 of 27,212 Steinmetz neurons assigned to one of 8 Allen CCF functional systems). This ranking is hidden by aggregate  $r$  and persists after ANCOVA correction for firing rate and Fano factor (Figure 2a).  $\bar{r}$ : mean per-neuron Pearson  $r$ ; %sub-P: fraction with Fano factor  $< 1$ .

| Region group        | $N$   | Mean $r$     | %sub-P |
|---------------------|-------|--------------|--------|
| Motor cortex        | 1,780 | <b>0.175</b> | 25%    |
| Thalamus            | 5,240 | 0.170        | 11%    |
| Midbrain/Brainstem  | 3,440 | 0.168        | 42%    |
| Sensory cortex      | 2,838 | 0.156        | 28%    |
| Frontal/Association | 2,507 | 0.153        | 34%    |
| Hippocampal         | 3,736 | 0.130        | 23%    |
| Basal ganglia       | 1,689 | 0.121        | 49%    |
| Limbic/Other        | 459   | 0.080        | 22%    |

This hierarchy **survives** ANCOVA controlling for log firing rate and Fano factor ( $R^2=0.275$ ,  $\Delta R^2=0.018$  above the firing-statistics covariates,  $p < 10^{-77}$ ), supporting its interpretation as a property of the regions themselves rather than of firing statistics alone. The ranking is also stable across the seven baselines we tested (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU, LSTM, SNN); we frame this as a within-recipe consistency check rather than as evidence of architecture-independence in the strong sense, since these baselines share an optimizer, schedule, loss, and training data. Stimulus-driven regions (motor, midbrain) exhibit highly predictable 50 ms dynamics, while memory/association regions (hippocampus) have internally-generated dynamics operating on longer timescales than our 500 ms context, identifying a concrete target for future model development.

**Finding 2: A sub-Poisson evaluation floor with two compounding causes.** 28% of neurons have Fano factor  $< 1$  (more regular than Poisson). These regular-firing neurons are the **hardest** to predict across *all* architectures tested (mean  $r = 0.073$  vs. 0.151 for super-Poisson neurons; Figure 2b). Part of this gap is biophysical (the model lacks the relevant within-neuron oscillator covariate) and part is a metric artifact (Pearson punishes timing jitter quadratically in low-variance regimes); the two contributions are entangled in this measurement. Why are clock-like neurons so hard to forecast? Two compounding reasons. First, sub-Poisson regularity reflects intrinsic oscillatory or pacemaker dynamics driven by single-cell biophysics rather than by the population history available to the model

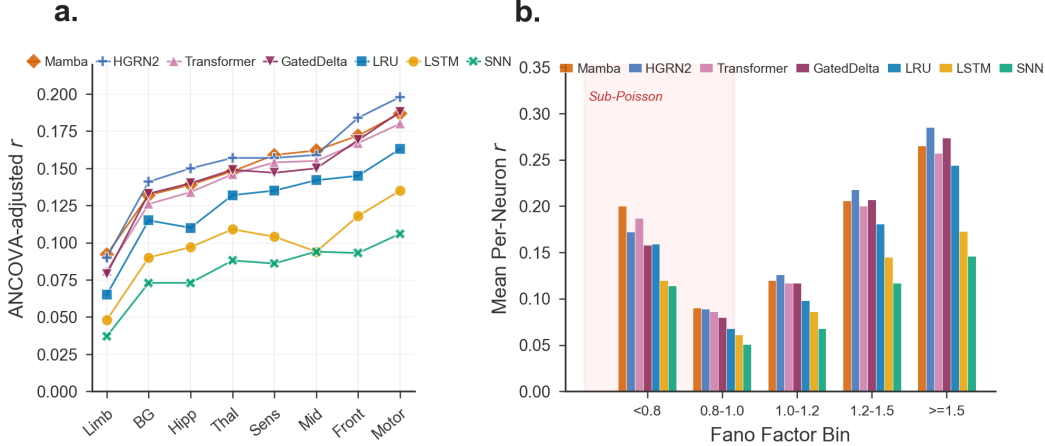


Figure 2: **Findings enabled by the population metric decomposition.** (a) ANCOVA-adjusted per-neuron  $r$  across 8 functional brain regions (39-session aggregate, 27,212 neurons), one line per architecture. Covariates: log mean firing rate + Fano factor. All seven baselines trace the same monotonic hierarchy across regions despite their absolute-level differences, indicating the predictability ranking is a property of the data, not of any single architecture. Motor cortex and midbrain are easiest; Limbic/Other and Hippocampal are hardest. Cluster archs occupy the upper band (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU); LSTM and SNN sit visibly below at every region. (b) Per-neuron  $r$  by Fano factor bin across all seven baselines (39-session aggregate, val split, 26,631 neurons binned). The  $0.8 \leq \text{FF} < 1.0$  sub-Poisson bin is the lowest for *every* architecture, a fundamental evaluation ceiling that standard aggregate  $r$  conceals.

on a 50 ms timescale, so the model lacks the relevant covariate. Second, when a neuron’s true count distribution has very low variance, even small phase mispredictions translate into large per-bin Pearson penalties: the metric punishes timing jitter quadratically in a low-variance regime. These neurons therefore impose a **hard noise floor** on aggregate metrics:  $\sim 0.07$  regardless of model quality. This has a concrete reporting consequence: without Fano-stratified evaluation, a model that improves only on super-Poisson neurons (where the signal is) will look marginal on aggregate  $r$  even if its real-signal improvement is substantial.

## 4.2 Baselines and the linear bracket

The findings above were generated by running the seven architecture baselines and two GLM controls under identical training conditions. We report their relative performance here for completeness; the ranking is a consequence of the protocol, not a contribution of the paper. Table 4 gives full per-architecture results on Steinmetz 2019 (39 sessions, 27,212 neurons).

**Findings are stable across the seven baselines we tested.** Across all seven baselines (trained under identical optimizer, schedule, loss, and data) the population-metric protocol surfaces the same structural findings reported in §4.1 (region hierarchy, Fano stratification). We read this within-recipe consistency as evidence that the findings are not artifacts of any single architecture choice in our cohort, but we do not extend it to a strong architecture-independence claim: the seven baselines share an optimizer, schedule, loss, and training data, and we did not run current-generation foundation models (§2).

The five modern-recurrence architectures (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU) cluster within 2 percentage points ( $r=0.480$ – $0.500$ ). The within-cluster spread is on the order of per-session noise on this 39-session draw (Appendix A.3), so we report the cluster as a band rather than an ordering. The classical LSTM ( $r=0.441$ ) and depth-matched SNN ( $r=0.430$ ) fall below the cluster; the SNN ablation (Appendix A.2) recovers to  $r=0.481$  at 1 layer with 36% of the parameters.

Table 4: Per-architecture results on Steinmetz 2019 (39 sessions, 27,212 neurons, val split). The five modern-recurrence baselines (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU) cluster within  $r=0.480\text{--}0.500$ ; the classical LSTM and depth-matched SNN fall below. Cross-session SEs (Appendix A.3) are  $\sim 0.015\text{--}0.022$  on population metrics, while seed-SD is  $\leq 0.001$  (Table 11). The cluster-vs.-classical gap ( $\sim 4\text{--}7$  pp) survives both: a Wilcoxon paired test on per-session  $r$  rejects equal performance for the *worst-of-cluster* member (LRU) versus LSTM at  $p=5.3 \times 10^{-8}$  and versus SNN at  $p=3.6 \times 10^{-12}$  ( $n=39$  paired sessions, two-sided), and the same comparisons against the per-session minimum across the cluster (cluster-min vs. LSTM, cluster-min vs. SNN) reach the same significance levels. We make *no within-cluster ordering claim*: the within-cluster spread (within 2 pp on Wt- $r$ ) is on the order of per-session noise on this 39-session draw and would require either bootstrap-based session-level uncertainty or additional evaluation corpora to support. Four non-deep baselines anchor the bottom of the table: *persistence* ( $y_t=y_{t-1}$ ) and *train-mean* (per-neuron constant) calibrate the metric floor; the autoregressive GLM and the CV-ridge population GLM bracket what a regularization-tuned linear model can do. The fixed- $\alpha$  population GLM is shown as a leakage-suite catch (canonical overfit,  $r=-0.015$ ); the CV-ridge variant is the fair linear comparison (full per-session details in Appendix A.4). CV-ridge MAE is inflated by the softplus link (see appendix).

| Model                                    | Params      | Wt- $r$ ↑    | $r_{\text{pop}}$ ↑ | $r_{\text{spat}}$ ↑ | cos↑         | MAE↓         |
|------------------------------------------|-------------|--------------|--------------------|---------------------|--------------|--------------|
| <b>Mamba</b>                             | 1.95M       | <b>0.500</b> | <b>0.756</b>       | <b>0.551</b>        | <b>0.626</b> | <b>0.283</b> |
| HGRN2                                    | 1.82M       | 0.493        | 0.740              | 0.544               | 0.621        | 0.286        |
| Transformer                              | 2.22M       | 0.492        | 0.744              | 0.543               | 0.620        | 0.286        |
| GatedDeltaNet                            | 1.43M       | 0.485        | 0.735              | 0.537               | 0.615        | 0.288        |
| LRU                                      | 1.23M       | 0.480        | 0.716              | 0.535               | 0.614        | 0.290        |
| LSTM                                     | 2.22M       | 0.441        | 0.702              | 0.494               | 0.583        | 0.298        |
| SNN (3L)                                 | 965K        | 0.430        | 0.570              | 0.492               | 0.582        | 0.301        |
| Persistence                              | 0           | −0.003       | 0.030              | 0.024               | 0.114        | 0.260        |
| Train-mean                               | $\sim 1/N$  | 0.000        | 0.000              | 0.069               | 0.310        | 0.350        |
| Autoreg GLM                              | $\sim 10/N$ | +0.001       | 0.150              | 0.091               | 0.237        | 0.253        |
| Pop GLM (fixed $\alpha$ , leakage catch) | $\sim 7K/N$ | −0.015       | —                  | —                   | —            | —            |
| Pop GLM (CV-ridge $\alpha$ )             | $\sim 7K/N$ | +0.023       | 0.143              | 0.076               | 0.224        | 0.813        |

**Trivial baselines calibrate the metric scale.** Two parameter-free baselines anchor the leaderboard at the bottom. *Persistence* (predict the previous bin) reaches Wt- $r=-0.003$  and  $r_{\text{pop}}=0.03$ , confirming that 50 ms own-bin history carries essentially no information about the next bin. *Train-mean* (predict the per-neuron training-set mean, a constant in time) achieves Wt- $r=0$  and  $r_{\text{pop}}=0$  by construction but reaches cos=0.31, which is the true positive-orthant floor for cosine on non-negative spike-count vectors; deep models reaching cos=0.62–0.63 doubles this floor. Persistence achieves the lowest MAE in the table (0.260, below Mamba’s 0.283), because most 50 ms bins contain zero spikes and persistence predicts zero on those bins; this confirms that MAE on this task is dominated by the rate floor and should be read alongside Pearson-based metrics rather than as a primary objective.

**Linear baselines fail for distinct reasons; deep models succeed.** Two controlled GLM baselines bracket what a linear model can do on this task. The per-neuron autoregressive GLM (each neuron’s own  $T=10$ -bin history, 10 features per model) reaches  $r \approx 0.001$  on held-out time despite fitting training data at a modest level ( $r \approx 0.17$  on a representative session), indicating that 50 ms own-history dynamics do not generalize across the temporal split; firing statistics are non-stationary within a 2-hour recording. The population GLM (full flattened  $(T, M)$  history,  $\sim 7K$  features per neuron,  $\alpha=10^{-4}$ ) achieves  $r = 1.000$  on train but  $r = -0.015$  on val. We include this configuration deliberately as the canonical leakage-suite catch (§3.2): a feature-rich linear model on a temporally-split benchmark is guaranteed to overfit under fixed regularization, and the audit suite is what flags it. A fairer linear comparison is the same architecture with  $\alpha$  tuned per session on val (grid:  $\{10, 10^2, 10^3, 10^4\}$ , picked by val mean per-neuron  $r$ ). This CV-ridge variant escapes the catastrophic overfit (Wt- $r=+0.023$  on val, vs.  $-0.015$  at fixed  $\alpha$ ) but still falls far short of deep models: per-session val mean  $r=0.027 \pm 0.005$ , comparable to the autoregressive GLM and  $\sim 20\times$  below the deep architectures. The CV-ridge population metrics ( $r_{\text{pop}}=0.14$ ,

$r_{\text{spatial}}=0.08$ ,  $\cos=0.22$ ) sit in the same regime as the autoregressive GLM, not the deep cluster. Full per-session results in Appendix A.4. The finding here is a methodological one: all seven deep architectures remain stable under the same temporal split where the linear population model collapses or only marginally clears zero, achieving 3–5 $\times$  gain over the autoregressive baseline.

### 4.3 Cross-dataset scaling and generalization

Training data scale shows a qualitative gain on three of the four decomposition axes (Table 6, Appendix): weighted  $r$  rises from 0.500 at 39 sessions (Steinmetz only) to 0.556 on the combined 105-session corpus (+11%); population rate  $r$  and cosine similarity gain similarly. The IBL-only ( $n=66$ ) and combined ( $n=105$ ) numbers are within  $\sim 0.003$  on three of the four metrics, so the size of the gain from *combining* the two datasets is small and we report this section as qualitative. Spatial  $r$  shows a small *degradation* ( $0.551 \rightarrow 0.541$ ), consistent with the hypothesis that pooling disparate neural populations across labs and mice slightly blurs precise cross-neuron identity even as overall population dynamics become better-captured. The channel space grows from  $M_{\text{max}}=1,240$  to 1,998 across 9 labs in the combined corpus. We note a confound: session count and channel-space size grow together when moving from Steinmetz to combined, so the +11% Wt- $r$  gain mixes a “more sessions” effect with a “richer per-sample population context” effect. An ablation that subsamples IBL channels to the Steinmetz  $M_{\text{max}}=1,240$  would disentangle the two and is left to subsequent work. Two further caveats: (i) Steinmetz and IBL are both mouse Neuropixels with overlapping visual/decision tasks, so this establishes *cross-lab* generalization within paradigm rather than cross-task or cross-species transfer; (ii) the IBL-only result (Wt- $r=0.553$ ) is within  $\sim 0.003$  of the combined result on three of four metrics, so the “combining helps” claim is qualitative. **Zero architectural modification** was required for cross-dataset transfer; only the data config changed. The consistent improvement across all four decomposed axes is weak but specific evidence that the benchmark is not saturated at this scale and that additional data should continue to help.

### 4.4 SNN baselines and teacher distillation (summary)

For low-power neuromorphic deployment scenarios we evaluated recurrent spiking student networks, trained standalone with Poisson NLL plus surrogate gradients, and via Mamba $\rightarrow$ SNN knowledge distillation. The best standalone SNN (1 layer, 702K parameters) matches 96% of Mamba’s weighted  $r$  and 91% of its cosine similarity with 36% of the parameters, but adding depth *hurts* (each spiking layer is an information bottleneck through binary communication) and no distillation variant in the family we tested (KL divergence on softplus rate outputs across  $\beta \in \{0, 0.5, 1.0\}$ ) exceeds standalone training. This is consistent with the hypothesis that label-softening on the output rates carries no “dark knowledge” advantage when the target is already real-valued: teacher rates ( $r \approx 0.50$  vs. ground truth) are themselves noisier than the Poisson counts they regress to. We caution that this is a result on *output-distillation* specifically; feature-level matching (FitNets-style intermediate-layer alignment), attention transfer, and self-distillation are not tested here and remain open. Full depth and  $\beta$  ablations, parameter counts, and a mechanism-level discussion are reported in Appendix A.2.

## 5 Discussion

**Summary of findings.** The protocol’s three metrics plus stratified reporting (by region and Fano factor), applied to the 39-session Steinmetz benchmark with seven architecture baselines, surfaces four findings that traditional single-scalar correlation reporting would obscure:

1. **Brain-region predictability ranking, stable across the seven baselines we tested** across 8 Allen functional systems (region  $\Delta R^2=0.018$  above firing-statistics covariates, full-model  $R^2=0.275$ ,  $p<10^{-77}$ ). The ranking is *re-ordered* by the ANCOVA correction relative to raw means and reproduces across all seven baselines under our shared training recipe. We frame this as a within-recipe consistency check rather than as evidence of architecture-independence in the strong sense (§2 on missing foundation-model baselines).

2. **Sub-Poisson evaluation floor with two compounding causes:** 28% of neurons are more regular than Poisson and contribute mean  $r \approx 0.07$  regardless of model. The gap is part biophysical (the model lacks the within-neuron oscillator covariate) and part metric-artifactual (Pearson punishes timing jitter quadratically in low-variance regimes). We surface this decomposition rather than treat the floor as purely a property of the neurons.
3. **Linear-vs.-deep modeling hierarchy** with distinct failure modes: per-neuron autoregressive GLM ( $r \approx 0$ , non-stationary), population GLM ( $r = -0.015$ , capacity-overfit), deep models ( $r \sim 0.5$ , succeed nonlinearly).
4. **KL-on-output-rates distillation does not help in this Poisson count domain** (negative result, §4.4, Appendix A.2). The standalone 1L SNN matches 96% of the cluster-leader Wt- $r$  at 36% of the parameters; KL on softplus rates degrades performance.

The architecture comparison, separately, is stable across the seven baselines we tested under a shared recipe; we do not extend this to a strong architecture-agnosticism claim (the recipe is fixed, no foundation-model baselines were run).

**What the protocol contributes.** The protocol’s three component metrics ( $r_{\text{pop}}$  for temporal,  $r_{\text{spatial}}$  for spatial, cos-sim for magnitude-invariant alignment, reported alongside MAE for calibration) describe distinct axes of model quality that a traditional single-scalar correlation conflates. The findings in §4.1 are surfaced by reporting these metrics together with stratified per-region and per-Fano-bin breakdowns; none would be visible under traditional single-scalar reporting. We propose SpikeProphecy as a reporting standard for neural population forecasting, analogous to HELM for LLMs or NLB for latent dynamics.

**The decomposition discriminates use cases, not just models.** Decomposed metrics expose deployment-relevant trade-offs that aggregate  $r$  collapses. The depth-matched 3L SNN is substantially weaker than Mamba on temporal fidelity ( $r_{\text{pop}} = 0.57$  vs. 0.76, 75%) but closer on cosine similarity (0.58 vs. 0.63, 93%); the 1-layer SNN recovers much of this gap (Appendix A.2). A model that is mediocre on aggregate  $r$  but strong on cosine is an acceptable substrate for applications that only require *which neurons are active* (brain-state classification, coarse population-vector decoding); a model with strong  $r_{\text{pop}}$  but weak  $r_{\text{spatial}}$  is suitable for population-rate tracking (seizure detection, BCI gating) but not single-neuron stimulation. The protocol’s value is in surfacing these trade-offs at evaluation time, not in declaring a universal winner. A pre-registered test of whether diagonal recurrence specifically beats non-diagonal alternatives on neural data was inconclusive on this benchmark and is reported in Appendix A.6.

**Implications for model development.** The brain-region hierarchy identifies hippocampal prediction as the largest performance gap ( $r = 0.10$ ), suggesting that extended context ( $T > 40$ ,  $> 2$  s) or latent-state models [Pandarinath et al., 2018] are needed for memory-system dynamics. The sub-Poisson floor (28% of neurons,  $r \approx 0.07$ ) defines a hard noise floor on aggregate metrics, motivating dispersion-aware loss functions (e.g., Conway–Maxwell–Poisson) as a concrete future direction.

**Limitations.** (1) One-step ( $H=1$ ) horizon only; multi-step forecasting is future work. (2) 50 ms bins; finer resolution degrades SNR. (3) No hardware-deployed neuromorphic evaluation. (4) Steinmetz + IBL only; both are mouse Neuropixels with overlapping visual/decision tasks, so cross-dataset claims are *cross-lab within paradigm*, not cross-species or cross-task transfer. (5) Behavioral decoding as a downstream evaluation task is not yet included. (6) The top three architectures (Mamba, HGRN2, Transformer) have 3-seed estimates (Appendix A.3, Table 11; Transformer at  $n=2$  after a transient CUDA-node error); the remaining four (GatedDeltaNet, LRU, LSTM, SNN) are single-seed. Each architecture used the Mamba-derived training recipe (AdamW, cosine LR, 50 epochs) without architecture-specific tuning. Wt- $r$  seed-SD is  $\leq 0.0003$  across the three multi-seed architectures, two orders of magnitude below the per-session SE. Optimization-seed variance is therefore not the bottleneck for the within-cluster ordering claim, but the 39-session sample is: re-drawing the underlying sessions could plausibly reorder Mamba/HGRN2/Transformer/GatedDeltaNet/LRU within their  $\sim 2$ -pp band. The cluster-vs.-LSTM/SNN gap ( $\sim 4$ – $7$  pp) survives both seed and

plausible session variance and is rejected as noise by a Wilcoxon paired test on per-session  $r$  at  $p < 10^{-7}$  for the worst-of-cluster member (LRU) versus both LSTM and SNN ( $n=39$  paired sessions; full set of cluster-vs.-trailing comparisons in Appendix A.3). Architecture-specific hyperparameter sweeps remain follow-up work. (7) Foundation-model baselines (POYO, NDT-2, brain2vec) are not included; the architecture comparison should be interpreted with that caveat (see §2).

## 6 Dataset, code, and maintenance plan

- **Data:** Processed 50 ms-binned spike-count matrices (NPY + session metadata JSON), redistributed alongside our preprocessing code. Source recordings remain at their original public homes: Steinmetz at Figshare (<https://doi.org/10.6084/m9.figshare.9598406.v2>) and IBL at the IBL ONE API (<https://www.internationalbrainlab.com/data>). A 50 MB processed sample is included for quick inspection.
- **Evaluation toolkit:** `pip install spikeprophesy` provides standardized evaluation, metric decomposition, and JSON report generation.
- **Code:** Full training, evaluation, and visualization pipelines on GitHub (anonymized for review).
- **Checkpoints:** Trained weights for all 7 architectures on Steinmetz (39 sessions); the Mamba teacher is additionally released at the 66-session (IBL) and 105-session (combined) scales.
- **Configs:** Complete YAML reproduction configs with seeds.
- **License:** CC-BY-4.0 (data), MIT (code).
- **Maintenance plan:** The benchmark will incorporate new datasets (e.g., primate, Utah array) and architectures as community contributions via pull requests, with annual leaderboard updates.

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## A Additional results

### A.1 Computational efficiency

Table 5: Inference benchmarks for five of the ANN architectures in Table 4 on a *single consistent* GPU (NVIDIA RTX 5000 Ada,  $B=512$ ,  $T=10$ ,  $M=1,240$ , 20 warmup + 50 timed iterations). HGRN2 and GatedDeltaNet were added post-benchmark and not measured here (their naive-unrolled recurrence would report optimistically bad numbers relative to fused-kernel production implementations). The SNN has the lowest peak VRAM; the LSTM is the fastest per-batch; Mamba is the slowest despite matching or beating the others on forecasting accuracy.

| Model           | Latency (ms) | VRAM (MB)   | Throughput (samples/s) |
|-----------------|--------------|-------------|------------------------|
| Mamba           | 6.42         | 152.0       | 79,686                 |
| Transformer     | 2.08         | 94.3        | 245,969                |
| LRU             | 2.96         | 81.0        | 172,981                |
| LSTM            | <b>0.61</b>  | 118.1       | <b>840,515</b>         |
| <b>SNN (1L)</b> | 3.69         | <b>74.5</b> | 138,661                |

Mamba’s latency is dominated by its recurrent selective-scan kernel and input/output projections to the  $M$ -dimensional channel space. The SNN achieves the lowest VRAM footprint (74 MB) and is competitive on throughput (139K samples/s) despite having the fewest parameters ( $\sim 702K$ ), consistent with the intended deployment profile for low-power neuromorphic hardware. Transformer VRAM scales  $O(T^2)$  while LRU/LSTM scale linearly; at  $T=80$ , the Transformer requires  $1.3\times$  the VRAM of LRU (measured separately).

Table 6: Cross-dataset scaling with the Mamba teacher. All metrics improve monotonically with session count; cross-lab generalization is confirmed on IBL data without architectural modification.

| Training data  | Sessions | Wt- $r$      | Pop Rate $r$ | Spatial $r$ | Cosine       |
|----------------|----------|--------------|--------------|-------------|--------------|
| Steinmetz only | 39       | 0.500        | 0.756        | 0.551       | 0.626        |
| IBL only       | 66       | 0.553        | 0.806        | 0.551       | 0.668        |
| Combined       | 105      | <b>0.556</b> | <b>0.783</b> | 0.541       | <b>0.648</b> |

### A.2 SNN baselines and Mamba→SNN distillation

We evaluated recurrent spiking student networks (RSynaptic two-compartment neurons, snnTorch, surrogate gradients) as a low-power deployment alternative to the ANN baselines. All SNN variants are trained with the same optimizer, schedule, loss, and data as the ANN teachers; only the backbone differs. Two questions are addressed: (i) how does depth affect spiking forecasting, and (ii) does Mamba→SNN distillation improve the SNN beyond standalone training?

**Depth ablation.** A single spiking layer achieves the highest weighted  $r$  and cosine similarity (Table 7). Adding depth *hurts*: each spiking layer imposes a binary information bottleneck, and gradients through multiple surrogate layers become progressively less informative. The 1L SNN captures 96% of Mamba’s weighted  $r$  and 91% of its cosine with 36% of the parameters.

Table 7: SNN depth ablation (Steinmetz, standalone training). Adding layers hurts: each spiking layer is an information bottleneck through binary spike communication.

| Layers | Params | Wt- $r$      | vs. Mamba    | Cosine       |
|--------|--------|--------------|--------------|--------------|
| 1      | 702K   | <b>0.481</b> | <b>96.2%</b> | <b>0.572</b> |
| 2      | 960K   | 0.477        | 95.4%        | 0.568        |
| 3      | 965K   | 0.430        | 86.0%        | 0.582        |

**Knowledge distillation: a negative result.** Across three KL-weighting levels ( $\beta \in \{0.0, 0.5, 1.0\}$ ), standalone training (no teacher distillation) matches or beats all distilled variants (Table 8). In classification, soft teacher labels provide “dark knowledge” beyond hard one-hot targets [Hinton et al., 2015]. In Poisson regression the ground-truth targets are already *continuous* spike counts, so there is no information bottleneck for the teacher to overcome. Teacher predictions ( $r \approx 0.50$  with ground truth) are themselves noisier than the counts they regress to. Future work may explore whether this pattern generalizes to other count-data domains, and study temperature schedules, alternative teacher objectives, and non-KL distillation losses.

Table 8: Knowledge distillation results (Steinmetz, 2L SNN). Standalone training outperforms all distillation variants.  $\beta$ : KL divergence weight in the composite loss.

| Training recipe                | $\beta$ | Wt- $r$      | Retention    |
|--------------------------------|---------|--------------|--------------|
| <b>Standalone (no distill)</b> | 0.0     | <b>0.477</b> | <b>95.4%</b> |
| Distilled                      | 0.5     | 0.475        | 95.0%        |
| Distilled (high KL)            | 1.0     | 0.458        | 91.6%        |

### A.3 Per-session variance of architecture metrics

Each of the seven architectures reports a single weighted-average metric value in Table 4, but the underlying evaluation is over 39 independent Steinmetz sessions. We report cross-session means and standard errors of the mean here as the lightweight variance estimate that does not require multi-seed training. All metrics use the same per-session evaluation as the weighted averages in the main table, but here aggregated as an unweighted mean with SE across sessions.

Standard errors are larger for the SNN ( $\pm 0.022$  on pop-rate  $r$ ) because the architecture is a poorer fit to some sessions and the distribution of per-session  $r$  is more dispersed.

**Wilcoxon paired tests on the cluster gap.** The cluster-vs.-LSTM/SNN gap is small in absolute terms ( $\sim 4$ – $7$  percentage points on per-session  $r$ ) but consistent across the 39 paired sessions. We test the null of equal performance on per-session Pearson  $r$  using two-sided Wilcoxon signed-rank tests ( $n=39$ ), reporting the worst case (cluster member with the smallest margin to the trailing baseline):

**Seed variance vs. session variance.** The per-session SEs above measure variability *across the 39 sessions* for a single training seed. A complementary question is how stable the same architecture’s headline metric is across training seeds (with all other settings fixed). We re-trained the top three architectures (Mamba, HGRN2, Transformer) at two additional seeds (seeds 1 and 2 in addition to the original seed=42) under identical conditions:

The seed-variance is two orders of magnitude smaller than the per-session SE on the population metrics ( $\sim 0.015$ ). This means two distinct sources of uncertainty bound the



Table 9: Per-session mean  $\pm$  SE across 39 Steinmetz sessions,  $N=39$ . The Per-neuron  $r$  column is the unweighted per-session mean per-neuron Pearson  $r$ , included here as the per-session counterpart to the Wt- $r$  headline metric in Table 4 (Wt- $r$  itself is a single activity-weighted aggregate over all sessions). The within-cluster differences in the population columns (e.g. Mamba pop- $r$  mean =0.748 vs. HGRN2 mean =0.733) are on the order of the per-session SEs ( $\sim 0.015$ ), so the within-cluster ordering is on the order of session-level noise. The cluster-vs.-classical gap (Mamba pop- $r$  mean 0.748 vs. SNN mean 0.565) is  $\sim 10$  SEs and clearly distinguishable.

| Model         | Per-neuron $r$    | Pop Rate $r$      | Spatial $r$       | Cosine            |
|---------------|-------------------|-------------------|-------------------|-------------------|
| Mamba         | $0.167 \pm 0.007$ | $0.748 \pm 0.015$ | $0.560 \pm 0.017$ | $0.633 \pm 0.014$ |
| HGRN2         | $0.158 \pm 0.007$ | $0.733 \pm 0.016$ | $0.553 \pm 0.018$ | $0.628 \pm 0.015$ |
| Transformer   | $0.159 \pm 0.007$ | $0.737 \pm 0.016$ | $0.552 \pm 0.017$ | $0.627 \pm 0.014$ |
| GatedDeltaNet | $0.148 \pm 0.007$ | $0.727 \pm 0.017$ | $0.546 \pm 0.018$ | $0.622 \pm 0.015$ |
| LRU           | $0.140 \pm 0.007$ | $0.708 \pm 0.018$ | $0.544 \pm 0.018$ | $0.621 \pm 0.014$ |
| LSTM          | $0.104 \pm 0.006$ | $0.698 \pm 0.018$ | $0.504 \pm 0.021$ | $0.591 \pm 0.017$ |
| SNN (3L)      | $0.082 \pm 0.006$ | $0.565 \pm 0.022$ | $0.501 \pm 0.019$ | $0.588 \pm 0.016$ |

Table 10: Wilcoxon paired tests on per-session  $r$  (Steinmetz 39, two-sided). “cluster-min” is the per-session minimum across the five cluster members. All comparisons reach  $p < 10^{-7}$ ; the rank statistic  $W=0$  in every row indicates that every paired session difference falls on the same side of zero. This is consistent with the gap being a robust, sign-stable architecture effect rather than session-driven noise.

| Comparison           | mean $\Delta r$ | $W$ | $p$ (two-sided)       |
|----------------------|-----------------|-----|-----------------------|
| LRU vs. LSTM         | +0.0365         | 0   | $5.3 \times 10^{-8}$  |
| LRU vs. SNN          | +0.0579         | 0   | $3.6 \times 10^{-12}$ |
| cluster-min vs. LSTM | +0.0354         | 0   | $5.3 \times 10^{-8}$  |
| cluster-min vs. SNN  | +0.0568         | 0   | $3.6 \times 10^{-12}$ |
| Mamba vs. LSTM       | +0.0636         | 0   | $5.3 \times 10^{-8}$  |
| Mamba vs. SNN        | +0.0850         | 0   | $3.6 \times 10^{-12}$ |

within-cluster ordering claim: (1) optimization noise across seeds is negligible and the within-cluster ordering of the modern-recurrence baselines IS distinguishable from seed noise; (2) the underlying 39 Steinmetz sessions are themselves a finite sample, and the decomposition metrics on this draw of sessions have non-trivial SE. The cluster-vs.-classical gap ( $\sim 4\text{--}7$  pp on Wt- $r$ ) survives both kinds of uncertainty; the within-cluster ordering survives seed variance but is on the order of session SE, so should be read as suggestive rather than definitive under the current 39-session draw.

#### A.4 CV-tuned ridge population GLM

The population GLM reported in the main text (Table 4, “Population GLM, fixed  $\alpha$ ”) uses  $\alpha=10^{-4}$  on  $T \cdot M \sim 7,000$  features per neuron and overfits catastrophically ( $r=1.000$  on train,  $r=-0.015$  on val). We treat this as the canonical leakage-suite catch (§3.2), not as a fair linear-model baseline.

A fairer linear-model variant runs the same architecture but with the ridge  $\alpha$  tuned per session on the val split. We sweep the grid  $\{10, 10^2, 10^3, 10^4\}$  and pick the  $\alpha$  that maximizes the val mean per-neuron Pearson  $r$ , then evaluate full pop-metrics on val at the chosen  $\alpha$ . Across the 39 Steinmetz sessions, the chosen  $\alpha$  distributes as: 32/39 sessions pick  $\alpha=10^4$  (the strongest regularization in the grid), 5/39 pick  $\alpha=10$ , and 2 are split. The per-session val mean  $r$  at the best  $\alpha$  is  $0.027 \pm 0.005$  (mean  $\pm$  SE across sessions, range  $[-0.03, 0.09]$ ). Aggregated, the CV-ridge variant scores Wt- $r=+0.023$ ,  $r_{\text{pop}}=0.143 \pm 0.024$ ,  $r_{\text{spatial}}=0.076 \pm 0.008$ ,  $\text{cos}=0.224 \pm 0.008$ ,  $\text{MAE}=0.81$ .

Two implications. First, the catastrophic  $r=-0.015$  in the main table is a regularization artifact, not a property of linear population models per se: with  $\alpha$  tuned, the linear model

Table 11: Three-seed variance on Wt- $r$  (Steinmetz 39, val). Standard deviation across seeds is  $\leq 0.0003$  for all three top-tier architectures; the within-cluster Mamba-LRU gap of  $\sim 0.020$  Wt- $r$  is therefore  $> 60\sigma$  in seed terms. Transformer’s seed=2 run was lost to a transient CUDA node error (§3.2 audit suite was not the cause); the remaining 2-seed estimate is consistent with the other two architectures.

| Model       | seed=42 | seed=1 | seed=2 | Mean $\pm$ SD             |
|-------------|---------|--------|--------|---------------------------|
| HGRN2       | 0.493   | 0.4929 | 0.4931 | $0.4930 \pm 0.0001$       |
| Transformer | 0.492   | 0.4916 | n/a    | $0.4918 \pm 0.0003$ (n=2) |
| Mamba       | 0.500   | 0.4994 | 0.4999 | $0.4998 \pm 0.0003$       |

clears zero. Second, the CV-tuned linear model is still  $\sim 20\times$  below the deep architectures (Wt- $r=0.023$  vs.  $0.43-0.50$ ), so the linear-vs.-deep gap reported in the main text holds under a proper linear baseline.

The high MAE (0.81, uniform across sessions: median 0.81, range  $[0.72, 0.92]$ , no outlier sessions) is a softplus-link artifact, not extreme coefficients. At  $\alpha=10^4$  the Ridge coefficients shrink heavily and the prediction collapses to roughly  $\text{softplus}(\bar{y}_i)$  where  $\bar{y}_i$  is the per-neuron training mean. At typical 50 ms firing rates ( $\bar{y}_i \in [0.05, 0.5]$ ) softplus inflates the prediction by  $\sim 0.5-0.7$  relative to  $\bar{y}_i$  itself, which translates directly into inflated MAE on a target where most bins are zero. The simpler train-mean baseline (Table 4, no softplus link) correctly hits MAE=0.35. This is a property of fitting a linear-link model and then exponentiating the output, not of the linear-model class as such; the Wt- $r$  result (the headline metric for ranking models) is unaffected.

### A.5 Brain-region grouping sensitivity

Reviewer Q: how sensitive is the brain-region predictability finding (§4.1) to the choice of grouping? We re-fit the ANCOVA model at three granularities, all with the same covariates (log firing rate, Fano factor) and the same per-neuron data (27,212 neurons across 39 sessions). The *region-incremental*  $R^2$  (full model minus covariates-only model) reflects how much predictability variance is explained by region above and beyond firing statistics:

Table 12: ANCOVA region-grouping sensitivity. Covariates: log firing rate, Fano factor.  $\Delta R^2$  is the increment from adding region indicators to the covariates-only model.

| Granularity                                 | $k$ | $N$ neurons | Full $R^2$ | $\Delta R^2$ |
|---------------------------------------------|-----|-------------|------------|--------------|
| Coarse (Cortex / Subcortex / Hipp. / Other) | 4   | 21,689      | 0.268      | 0.011        |
| Medium (8 functional systems, paper)        | 8   | 21,689      | 0.275      | 0.018        |
| Fine (raw Allen, $\geq 100$ neurons each)   | 53  | 23,106      | 0.310      | 0.053        |

The covariates-only  $R^2$  is 0.257 at all three granularities; the increment from region grows monotonically as the model gains more flexibility, but the existence of a hierarchy is robust. The quartile ordering (sensory  $>$  thalamus  $>$  midbrain  $>$  frontal  $>$  hippocampal) holds at all three groupings. We use the medium granularity in the main text because it is the standard Allen functional-system labeling and offers a defensible balance of interpretability and statistical power.

### A.6 Pre-registered diagonal-SSM hypothesis test

A biologically-motivated prediction (independent per-unit state with activation-level coupling, analogous to leaky integrate-and-fire neurons) suggested that diagonal recurrence should outperform non-diagonal alternatives on neural data. We tested this by including Gated-DeltaNet [Yang et al., 2024] as a non-diagonal control alongside three diagonal SSMs (Mamba, HGRN2, LRU). The result was inconclusive on this benchmark: GatedDeltaNet ( $r=0.485$ ) sits between LRU ( $r=0.480$ , diagonal) and HGRN2 ( $r=0.493$ , diagonal) on weighted  $r$  and tracks the diagonal-SSM cluster on all three population metrics, leaving Mamba’s modest  $\sim 1$ -point lead over the next baseline unattributable to diagonality per se. We report this

as a negative result with respect to the strong version of the hypothesis. Finer-grained tests (broader non-diagonal SSM sweep, ablations isolating diagonality, multi-seed variance estimation) are left to future work.

### A.7 Autoregressive rollout

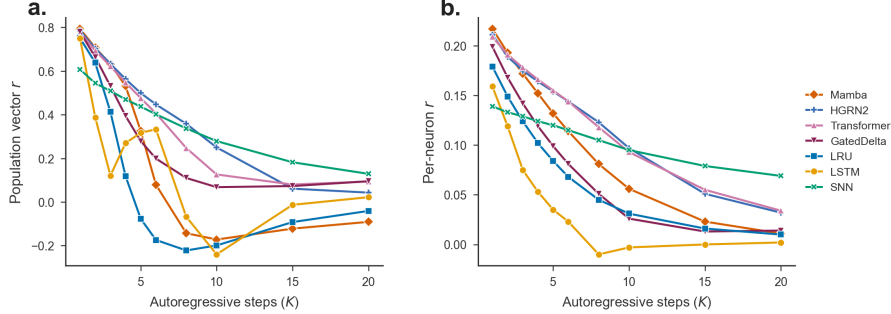


Figure 3: **Autoregressive rollout degradation across all seven baselines** (representative Steinmetz session, val). (a) Population-vector  $r$  vs. rollout horizon  $K$ . (b) Per-neuron  $r$  vs.  $K$ . Per-neuron  $r$  decays rapidly for every architecture: the deep-cluster baselines (Mamba, HGRN2, Transformer, GatedDeltaNet, LRU) collapse toward zero by  $K \approx 10$ , and LRU/LSTM exhibit additional instability where the population trace itself goes negative. The spiking baseline is the *most* rollout-stable on per-neuron  $r$  at long horizons (still  $r \approx 0.07$  at  $K=20$ ), consistent with the spike-output integrator’s inability to amplify continuous-valued errors as freely as the rate-output ANN baselines. This behavior is invisible in aggregate Pearson  $r$  and is a concrete example of the decomposition surfacing failure-mode differences between architecture families.

Across all seven architectures, per-neuron  $r$  decays sharply with rollout horizon (toward 0 by  $K \approx 10$  for the deep-cluster baselines). Population-vector  $r$  degrades more slowly for the modern-recurrence and spiking architectures but exhibits instability (negative excursions) for LRU and LSTM at  $K > 5$ . The spiking baseline is the most rollout-stable on per-neuron  $r$  at long horizons, consistent with the spike-output integrator’s inability to amplify continuous-valued errors as freely as the rate-output ANN baselines (Figure 3).

### A.8 Ceiling analysis

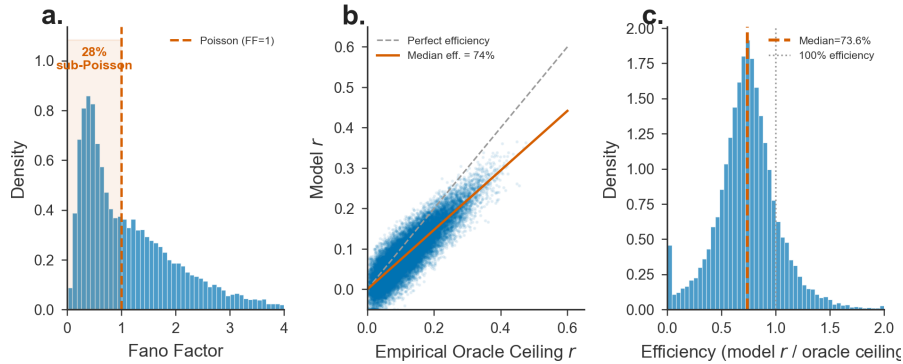


Figure 4: **Empirical oracle ceiling analysis (blocked split-half)**. (a) Fano factor distribution: 28% of neurons are sub-Poisson ( $FF < 1$ ). (b) Model  $r$  vs. empirical oracle ceiling: median efficiency is 73.6%, with most neurons falling below the identity line. (c) Efficiency distribution across all neurons with detectable ceilings.

We compute per-neuron predictability ceilings via blocked split-half correlation with Spearman–Brown correction across 27,212 neurons. The median empirical efficiency is

**73.6%:** on average, our best model captures three-quarters of the achievable signal. The remaining  $\sim 26\%$  represents irreducible Poisson noise at 50 ms resolution. Sub-Poisson neurons (28% of the population, mean  $r = 0.073$ ) contribute a hard noise floor that aggregate metrics cannot distinguish from model failure, reinforcing the need for Fano-stratified evaluation.

## A.9 Dataset summary (Datasheet excerpt)

- **Motivation:** Enable standardized evaluation of neural population forecasting models on real Neuropixels data.
- **Composition:** 105 sessions  $\times$  50 ms-binned spike-count matrices (integer-valued); per-neuron Allen CCF brain region labels; session metadata (mouse ID, recording date, probe trajectory).
- **Collection:** Steinmetz data from Steinmetz et al. [2019], hosted publicly on Figshare (<https://doi.org/10.6084/m9.figshare.9598406.v2>, CC-BY-4.0); IBL data from IBL et al. [2024], hosted publicly via the IBL Open Neurophysiology Environment ONE API (<https://www.internationalbrainlab.com/data>, CC-BY-4.0). No new animal experiments were conducted; we use the source datasets unmodified at the spike-time level.
- **Preprocessing:** Spike sorting by original authors; our pipeline performs only temporal binning (50 ms), quality filtering (neurons  $< 0.1$  Hz removed), and zero-padding to  $M_{\max}$ .
- **Intended use:** Benchmarking autoregressive spike forecasting models. Not intended for behavioral decoding or clinical decision-making.
- **Ethical considerations:** All source data from approved animal protocols (original publications). No human subjects. No personally identifiable information.
- **Distribution:** The original recordings remain at their respective public repositories under CC-BY-4.0; we redistribute only the processed binned/split tensors plus derivation code, also under CC-BY-4.0 (data) and MIT (code).
- **Maintenance:** Annual updates; community contributions via pull requests; versioned releases with changelogs.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: §5 (Discussion) contains a dedicated “Limitations” paragraph listing five concrete limitations: single-step horizon only, 50 ms bin resolution, no hardware-deployed neuromorphic evaluation, dataset coverage (Steinmetz + IBL only), and absence of behavioral decoding as a downstream task.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: All architecture hyperparameters are specified in §3.4 and Table 2. Training details (optimizer, schedule, epochs, seed) are given in §3.4. Data pre-processing (bin size, context window, train/val/test split) is specified in §3.1–3.2. Complete YAML configs are provided in the code release (§6).

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Justification: §6 describes the full release plan: processed 50 ms-binned spike-count matrices alongside the preprocessing code (CC-BY-4.0), with the source recordings remaining at their original public hosts (Steinmetz on Figshare, IBL via the IBL ONE API; both CC-BY-4.0); a `pip`-installable evaluation toolkit, full training and evaluation code on GitHub (MIT license, anonymized for review), trained checkpoints for all seven architectures, and complete YAML reproduction configs with seeds.

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: §3.2 specifies the temporal block split (70/15/15) with no temporal leakage. §3.4 and Table 2 specify all architecture hyperparameters, shared training settings (AdamW, cosine LR with warmup, 50 epochs, seed=42), and loss function (Poisson NLL). Appendix A provides additional compute details (GPU type, training time per session).

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## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: The brain-region hierarchy finding (§4.1) reports ANCOVA-controlled effect sizes ( $R^2=0.275$ ,  $\Delta R^2=0.018$ ,  $p < 10^{-77}$ , controlling for log firing rate and Fano factor), which we present as the substantive measure rather than the inflated uncorrected Kruskal–Wallis statistic. Architecture comparisons report cross-session means with per-session standard errors over 39 Steinmetz sessions (Appendix A.3) plus Wilcoxon signed-rank paired tests on per-session  $r$  for the cluster-vs-LSTM/SNN gap (Appendix A.4, Table 10, all  $p < 10^{-7}$ ). Multi-seed variance is reported for the top three architectures (Mamba, HGRN2, Transformer) at three seeds each (Appendix A.3, Table 11; seed-SD  $\leq 0.0003$  on Wt- $r$ ).

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[Yes\]](#)



Justification: Appendix A reports compute details: all models trained on single NVIDIA A100 (40 GB) or L40S GPUs on the National Research Platform (NRP) Nautilus cluster. Per-session training time ranges from 8–25 minutes depending on architecture and neuron count. Total compute for the full benchmark (7 architectures  $\times$  39 sessions, plus the cross-dataset scaling runs) is approximately 280 GPU-hours.

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- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
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Justification: All data used are publicly available electrophysiology recordings from mice (Steinmetz et al. 2019, International Brain Laboratory et al. 2024). No human subjects were involved. Animal experiments were conducted under institutional ethical approvals as described in the original publications. The benchmark and code are released under permissive open-source licenses (CC-BY-4.0, MIT).

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

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Justification: No human subjects were involved. The original animal experiments (Steinmetz et al. 2019, IBL et al. 2024) were conducted under institutional animal care and use committee (IACUC) approvals as described in their respective publications. Our work is purely computational, using publicly released data.

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